SOLUTIONS

- Types of Solutions
- Concentration of Solution
- Solubility
- Raoult's law
- Colligative Properties
- Vant Hoff factor

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1-2 Q/s

Types of Solutions

Table 2.1: Types of Solutions

Type of Solution	Solute	Solvent	Common Examples
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

Concentration of Solution

1. Mole fraction = $\frac{\text{mole of compound}}{\text{Total moles}}$

2. Molality = $\frac{\text{wt of solute} \times 1000}{\text{mol wt of solute} \times \text{wt of solvent (g)}}$

3. Molarity = $\frac{W_{solute} \times 1000}{\text{mol wt} \times \text{Volume (ml)}}$

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4. % weight by volume

5. % weight by mass

6. Normality = $\frac{\text{Equivalent of solute}}{\text{Vol. of solution (L)}}$

Molarity, Normality : Temperature Dependent
Molality, Mole fraction : Temperature Independent

Solubility of Gases (Henry's Law)

Solubility of gas $\propto$ Pressure of gas at surface of liquid

Henry's Law

$$P = K_H x$$

K_H : Henry's Constant
x : mole fraction

Value of Henry's Constant depends upon nature of gas and temperature

More the value of K_H : lesser the solubility of gas

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Raoult's Law

Non Volatile Solute in Volatile Solvent

$$\frac{p^\circ - p_s}{p^\circ} = \chi_{solute}$$

p° = vapour pressure of pure solvent
p_s = vapour pressure of solution
χ_{solute} = mole fraction of solute

Volatile Solute in Volatile Solvent

$$p_A = p_A^\circ \chi_A$$

$$p_{total} = p_A + p_B$$

$$p_B = p_B^\circ \chi_B$$

p_A = vapour pressure of A (solvent)
p_B = vapour pressure of B (solute)

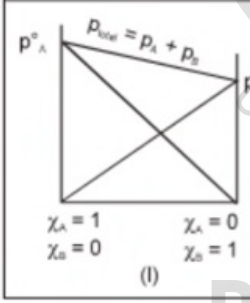
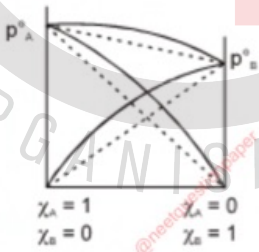
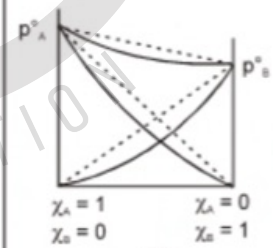
p°_A = vapour pressure of pure A
p°_B = vapour pressure of pure B

χ_A and χ_B = mole fractions of A & B

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Ideal & Non Ideal Solution

The solutions which obey Raoult's law are ideal solutions and those which do not obey Raoult's law form non-ideal solutions.

Ideal Solution	Non-Ideal Solution	
	Positive Deviation	Negative Deviation
1. Obey's Raoult's law	1. Disobey's Raoult's law	1. Disobey's Raoult's law
2. $p_A = p^\circ_A \chi_A$ $p_B = p^\circ_B \chi_B$ $p_{total} = p_A + p_B$	2. $p_A \neq p^\circ_A \chi_A$ $p_B \neq p^\circ_B \chi_B$ $p_{total} \neq p_A + p_B$ $[P_{total} > P_A + P_B]$	2. $p_A \neq p^\circ_A \chi_A$ $p_B \neq p^\circ_B \chi_B$ $p_{total} \neq p_A + p_B$ $[P_{total} < P_A + P_B]$
3. $\Delta H_{mix} = 0$ $\Delta V_{mix} = 0$	3. $\Delta H_{mix} = +ve$ $\Delta V_{mix} = +ve$	3. $\Delta H_{mix} = -ve$ $\Delta V_{mix} = -ve$
4. Interaction A – B = A – A/B – B *e.g.* Chlorobenzene + Bromobenzene	4. Interaction A – B < A – A/B – B *e.g.* $CH_3OH - H_2O$	4. Interaction A – B > A – A/B – B *e.g.* $CH_3COCH_3 + CHCl_3$
 (I)	 (II)	 (III)

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Colligative Properties

Properties that depend on number of solute particles dissolved in it.

1. Relative Lowering of Vapour Pressure

$$\frac{P^\circ_A - P_A}{P^\circ_A} = x_B = \frac{w_2}{M_2} \times \frac{M_1}{w_1}$$

2. Osmotic Pressure

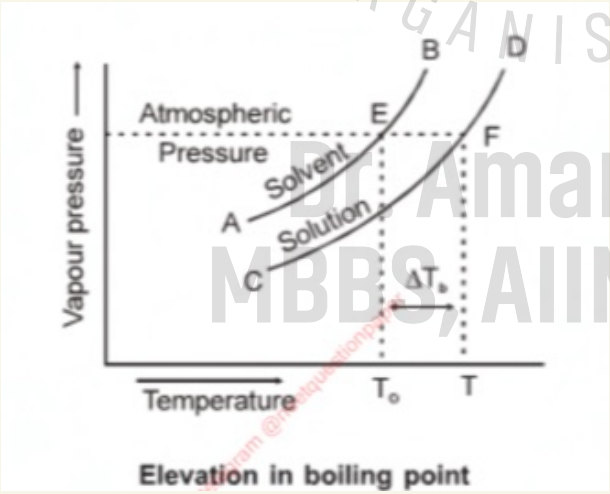
$$\pi = CRT$$

$$\pi = \frac{n}{V} RT$$

(C : concentration)

Osmotic pressure is excess pressure must be applied on a solution to prevent passage of solvent into it through semipermeable membrane.

3. Elevation in Boiling point



Elevation in boiling point

$$\Delta T_b = T - T_0$$

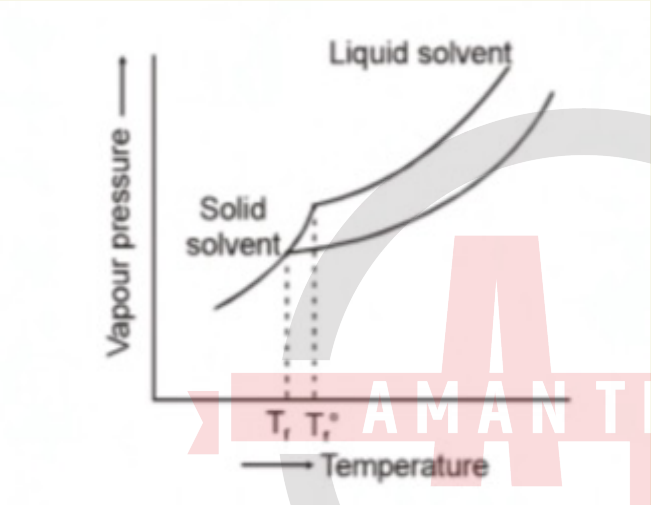
$$\Delta T_b = K_b m$$

(m = molality)

K_b : molal elevation constant or Ebullioscopic constant

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4. Depression in Freezing point



$\Delta T_f = K_f m$

K_f : Molal Depression Constant OR Cryoscopic Constant

K_b and K_f : Intensive Properties of Solvent

Van't Hoff factor (i)

$$i = \frac{\text{Observed value of Colligative Property}}{\text{Theotrical value}}$$

$$i = \frac{\text{no. of moles after dissociation}}{\text{no of moles before dissociation}}$$

$i = 1$	Ideal solution
$i > 1$	Dissociation
$i < 1$	Association

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Degree of Association

$$\alpha = \frac{\text{No. of moles associated}}{\text{Total No. of moles taken}}$$

$$\alpha = \frac{i-1}{\frac{1}{n}-1}$$

Degree of Dissociation

$$i = 1 + (n-1)\alpha$$

n : Total No. of ions produced per molecule of electrolyte

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PREVIOUS YRS IMPT

Which of the following statements is correct regarding a solution of two components A and B exhibiting positive deviation from ideal behaviour? [NEET-2019(Odisha)]

A. Intermolecular attractive forces between A-A and B-B are equal to those between A-B.

B. Intermolecular attractive forces between A-A and B-B are stronger than those between A-B.

C. $\Delta_{mix}H = 0$ at constant T and P.

D. $\Delta_{mix}V = 0$ at constant T and P.

The correct answer is 'B'

UNDERLYING CONCEPT

- For positive deviation from ideal solution,

$\Delta_{mix}H > 0$ and $\Delta_{mix}V > 0$

At 100°C the vapour pressure of a solution of 6.5 g of a solute in 100 g water is 732 mm. If $K_b = 0.52$, the boiling point of this solution will be [NEET-2016]

A. 103°C

B. 101°C

C. 100°C

D. 102°C

ANSWER AND EXPLANATION

$P° = 760$ mm (At 100°C)

$$\frac{P^\circ - P_s}{P_s} = \frac{n_2}{n_1} \Rightarrow \frac{760-732}{732} = \frac{n_2}{100/18} \Rightarrow n_2 = 0.2$$

$\Delta T_b = K_b.m \Rightarrow T_b - T_b° = K_b \frac{n_2 \times 1000}{W_A}$

$\Rightarrow T_b - 100°C = \frac{0.52 \times 0.2 \times 1000}{100} = 1.04$

$T_b = 101.4 \approx 101°C$

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p_A and p_B are the vapour pressure of pure liquid components, A and B, respectively of an ideal binary solution. If x_A represents the mole fraction of component A, the total pressure of the solution will be [AIPMT (Prelims)-2012]

A. $p_B + x_A (p_B - p_A)$

B. $p_B + x_A (p_A - p_B)$

C. $p_A + x_A (p_B - p_A)$

D. $p_A + x_A (p_A - p_B)$

ANSWER AND EXPLANATION

Using Raoults law

$$P = p_A x_A + p_B x_B$$
$$= p_A x_A + p_B (1 - x_A)$$
$$= p_A x_A + p_B - p_B x_A)$$
$$= p_B + (p_A - p_B) x_A$$

Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct? Assume that the temperature is constant at 25°C. (Given, Vapour Pressure Data at 25°C, benzene = 12.8 kPa, toluene = 3.85 kPa) [NEET-2016]

A. Not enough information is given to make a prediction

B. The vapour will contain a higher percentage of benzene

C. The vapour will contain a higher percentage of toluene

D. The vapour will contain equal amounts of benzene and toluene

ANSWER AND EXPLANATION

$$x_{Benzene} = x_{Toluene} = \frac{1}{1+1} = 0.5$$

$$y_{Benzene} = \frac{P^{\circ}_{Benzene} \cdot x_{Benzene}}{P^{\circ}_{Benzene} x_{Benzene} + P^{\circ}_{Toluene} x_{Toluene}}$$

$$= \frac{12.8 \times 0.5}{12.8 \times 0.5 + 3.85 \times 0.5} = 0.77$$

Thus, the vapour will contain higher percentage of benzene.

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